

Patrick Kariuki

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EDUCATION

Johns Hopkins University

Baltimore, MD

Bachelor of Science in Computer Science

Expected May 2026

- **Relevant Coursework:** Gateway Computing (Java), Intermediate Programming (C/C++), Data Structures, Algorithms, Full-Stack JavaScript, Artificial Intelligence, Computer Systems Fundamentals, Introduction to Data Science, Object-Oriented Software Engineering, Databases, Web Security.

EXPERIENCE

Course Assistant

August 2024 – Present

Johns Hopkins University

Baltimore, MD

- Mentored Intermediate Programming students with focus in object-oriented programming with C++ and supported learning through office hours.
- Served as project manager for team of six students in Practical Generative AI course, guiding development of an applied half-semester project with structured check-ins and feedback cycles.
- Coordinated study resources and streamlined grading processes in collaboration with faculty and teaching assistants.

Tech Developer Intern

May 2024 – August 2024

Sponsors for Educational Opportunity (SEO)

New York, NY

- Developed StarSight, a full-stack web application that provides real-time stargazing predictions by integrating weather data, light pollution APIs, and astronomical databases.
- Built scalable Flask backend with RESTful API endpoints, implementing JWT authentication, secure session management, and PostgreSQL database integration for user preferences and location data.
- Trained constellation recognition model using Principal Component Analysis on astronomical datasets allowing users to upload night sky images and receive constellation prediction with 85% accuracy.
- Automated deployment pipeline using Ansible for configuration management and Docker, enabling consistent deployments across development and production environments.

PROJECTS

LeetScribe | *React, Chrome Extension API, Convex, GitHub API*

June 2025 – Present

- Developed Chrome extension that automatically captures LeetCode solutions and generates UMPIRE-format documentation using AI integration.
- Implemented GitHub API integration for seamless solution syncing and version control of coding practice notes.
- Built structured documentation system that transforms raw code solutions into organized study materials for technical interview preparation.

Studia | *React, Python Flask, MongoDB*

January 2025 – May 2025

- Built full-stack AI-powered study assistant that generates adaptive explanations, quizzes, and flashcards.
- Integrated vector search using OpenAI embeddings to retrieve relevant content from Wikipedia and YouTube for supplemental learning, and Notion for note taking.
- Implemented multi-modal features including file uploads, speech-to-text, and text-to-speech for accessibility.

Flashcard App | *TypeScript, React, SQLite, Hono.js*

October 2024 – December 2024

- Engineered a full-stack flashcard application with React frontend and Hono.js REST API backend, featuring user authentication, deck management, card creation, and study functionality.
- Designed RESTful API endpoints with comprehensive error handling, input validation using Zod, and SQLite database integration via Drizzle ORM.
- Containerized frontend and API services using Docker and automated deployment through GitHub Actions.

TECHNICAL SKILLS

Programming: Java, Python, C, C++, SQL, JavaScript/TypeScript, HTML/CSS, Kotlin, Swift

Frameworks: React, Node.js, Flask, Django, FastAPI

Developer Tools: Git, Docker, Ansible, Visual Studio, Android Studio, Xcode

Leadership: President, Software Engineering Club (May 2025 - Present)